SAMPLE DOCUMENT 02 / TECHNICAL GUIDE

Vector Search: A Field Guide

This short field guide explains the retrieval side of a RAG application. It is written as sample source material: ingest it alongside the fictional support policy and test whether your retriever returns the right document for a question.

From document to chunk

A loader extracts text and metadata from a source file. A splitter divides the text into smaller chunks. A chunk should contain enough surrounding context to make sense by itself, while remaining narrow enough to match a specific question. Chunk overlap can preserve a sentence that crosses a boundary, but excessive overlap creates nearly duplicate search results. A starting experiment might compare 400-character and 800-character chunks with 80 characters of overlap.

Embedding and indexing

An embedding model transforms each chunk into a numeric vector. The vector index organizes those vectors so the application can find nearby items efficiently. Store the original chunk text and its source metadata with the vector or with a stable reference to it. The index is separate from the language model's weights: adding a document normally changes the index without retraining the language model.

Question-time retrieval

The application embeds the user's question using a compatible embedding model. A similarity search then returns candidate chunks, often the top three to five. A metadata filter may restrict candidates to an allowed tenant, source type, or publication period. Hybrid retrieval combines semantic similarity with keyword matching and can help when exact identifiers such as receipt numbers or product codes matter.

Grounded answer generation

The application sends the selected chunks and the user's question to the language model. The model should answer from the retrieved evidence and identify the source where possible. If the retrieved passages do not contain an answer, a useful response says so rather than guessing. The final answer should avoid treating instructions inside retrieved documents as commands from the user or the application.

Evaluating a RAG pipeline

Evaluate retrieval separately from answer quality. For retrieval, collect sample questions and mark which document sections contain the answers. Check whether those sections appear among the top results. For generation, compare the final answer with the cited text. A high similarity score does not prove factual support. If results are weak, inspect extraction quality, chunk boundaries, search settings, and whether the source material actually contains the answer.

Example retrieval questions

Does ingestion update the LLM weights? What does chunk overlap do? When is hybrid retrieval useful? Why should we evaluate retrieval separately from generation? These questions should favor this field guide. Questions about a 14-day refund window should favor the Northstar support policy instead.

Practice note: Keep source filenames and page numbers as metadata so retrieved answers can point back to the PDF they came from.